QP Code: 24900024

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

FIRST SEMESTER MGU-UGP (HONOURS) REGULAR EXAMINATION
NOVEMBER 2024

Discipline Specific Core Course-MGIDSCBCH100- BIOCHEMISTRY THE SCIENCE OF LIFE

(2024 Admission Onwards)

Scheme for Valuation

Part A
Multiple Choice Questions
Answer all questions. Each question carries 1 mark

1. Which of the following heteropolysaccharides play a role in blood clotting? [U][2]
   Answer: Option b-Heparin
2. The secondary structure of protein is stabilized by [U][3]
   Answer: Option d-All of these
3. What type of bond links nucleotides in a DNA strand? [K][4]
   Answer: Option c- Phosphodiester linkage
4. Cholesterol is categorized as which type of lipid? [An]][4]
   Answer: Option c- Steroid
5. In pure water, the concentrations of H+ and OH- are [U][1]
   Answer: Option b- Equal and both are 1.0 x 10^{-7} M (5 x 1= 5)

Part B
Fill in the Blanks
Answer all questions. Each question carries 2 marks

6. In heteropolysaccharides, the monosaccharide units are linked by……. bonds. [U][2]
   Answer: Glycosidic
7. The …… form of DNA is the most stable and common form under physiological [U][4]
   Conditions **Answer: B form**
8. A buffer solution is a mixture of a weak acid and its …. [U][1]
   Answer: Conjugate base
9. B DNA has approximately ….base pairs per turn of the helix [U][4]
   Answer: 10
10. ……is the storage form of glucose in animals. [U][2]
    Answer: Glycogen (5 x 2= 10)

Part C
Short Answer Questions
Answer any 5 questions. Each question carries 3 marks

11. Evaluate the role of phosphodiester linkages in the formation and stability of nucleic acid structures. [E][4]
    Answer: Description of phosphodiester linkage- **1 mark**
    Structural representation -**1 mark**
    Stability/ Directionality of DNA strands -**1 mark**

12. Draw the structure of water. [K][1]
    Answer: Draw the structure of water-**1 mark**
    Indicate charge separation- **1 mark**
    Specify the bond angle-**1 mark**
13. What are the two main components of starch, and how do they differ in structure? [U][2]
    Answer: Mention amylose and amylopectin -**1 mark**
    Structural description of both- **1 mark each**
14. How does the bicarbonate buffer system help regulate the pH in blood plasma? [K][1]
    Answer: Mention proton donor and acceptor **-1 mark**
    Low pH- bicarbonate ions (HCO_3^-) react with hydrogen ions to form carbonic acid (H_2CO_3). High pH- carbonic acid dissociate into hydrogen ions and bicarbonate, releasing hydrogen ions to lower the pH- **1 mark**
    $\frac{[HCO_3^-]}{[H_2CO_3]}$= 20- **1 mark**
15. Discuss the relationship between pH and pOH in an aqueous solution? [An][1]
    Answer: Definition/explanation of pH, pOH **1.5 marks**
    pH + pOH = 14 at 25°C **1.5 marks**
16. Explain epimerization with suitable examples? [U][2]
    Answer: Definition of epimerization- **1mark**
    Structures of Glucose, Galactose/Mannose -**2 marks**
17. What is a zwitterion? [K][3]
    Answer: Definition of zwitterion-**1 mark**
    Structure of zwitterion-**2 marks** (5 x 3= 15)

Part D

Short Essay type questions

Answer any 4 questions. Each question carries 5 marks

18. What are heteropolysaccharides, and how do they differ from homopolysaccharides?[U][2]
    Answer: Heteropolysaccharides with example -**2 marks**
    Homopolysaccharide with example -**2 marks**
    Structural details of any 1 heteropolysaccharide-**1 mark**
19. Discuss the contributions of James Watson and Francis Crick to our understanding of [An][4]
    the DNA structure.
    Answer: Detailed explanation of Watson and Crick's double helix model.-**2 marks**
    Illustration of double helix-**2 marks**
    Mention the use of X-ray diffraction data and Chargaff rule -**0.5 mark each**
20. Describe the structural organization of proteins [U][3]
    Answer: Primary structure-**1 mark**
    Secondary structure-**1 mark**
    Tertiary structure-**1 mark**
    Quaternary structure-**1 mark**
    Schematic representation of 4 levels of organization-**1 mark**
21. Describe the structure of lecithin (phosphatidyl choline). How are its components [U][4]
    organized, and what role does each component play in its function?
    Answer: Structure of Lecithin-**3 marks**
    Function-**2 marks**

22. Explain what is Biochemistry, and why is it considered a pivotal field in the life sciences? [U][1]

Answer: Definition of Biochemistry-**1 mark**
Bridges chemical and biological sciences-**2 marks**
Understanding of biomolecules and chemical basis of life processes-**2 marks**

23. Discuss in detail about a) stereoisomers b) epimers c) anomers. [U][2]

Answer: Stereoisomers-**1 mark**
Epimers-**2 marks**
Anomers-**2 marks**
Examples with structures. (4 x 5= 20)